Question	Answer	Marks	Partial Marks
1(a)(i)	$\frac{450}{8+7+3}\times 8$ oe	**2**	**M1** for $\frac{450}{8+7+3}$
1(a)(ii)	75	**1**	
1(a)(iii)	56	**2**	**M1** for $\frac{32}{100}\times(450 - 200 - \textit{their } 75)$ oe or $\frac{32}{100}\times\frac{450}{8+7+3}\times 7$ oe If 0 scored, **SC1** for answer 231
1(a)(iv)	59 000 nfww	**3**	**B2** for 58 600 to 58 800 or **B1** for 293 to 294 or **M1** for $\frac{\text{figs}485\times 200}{165}$ oe If 0 scored, **SC1** for *their* more accurate answer seen and rounded to the nearest 1000
1(b)(i)	3 075 000	**1**	
1(b)(ii)	3.075×10^6	**1**	**FT** *their* (b)(i)
1(c)	32.5	**2**	**M1** for $x\times\left(1+\frac{16}{100}\right) = 37.7$ or better
1(d)	2460 or 2458. …	**2**	**M1** for $1800\left(1+\frac{2.1}{100}\right)^{15}$ oe
2(a)(i)	90	**1**	
2(a)(ii)	68	**1**	
2(a)(iii)	52	**1**	**FT** 120 – *their* (a)(ii)
2(a)(iv)	20	**2**	**B1** for 60 in working or as answer
2(b)(i)	97.5	**4**	**M1** for mid-points soi (50, 70, 90, 115, 145, 180) **M1** for use of Σfm with m in correct interval including both boundaries **M1** for (dep on 2nd M1) for $\Sigma fm \div 80$
2(b)(ii)	Bars with heights 0.9, 0.5, 0.3, 0.175 and with correct widths	**4**	**B1** for each correct bar If 0 scored, **SC1** for 3 or 4 correct frequency densities

Question	Answer	Marks	Partial Marks
2(b)(iii)	$\frac{28}{395}$ oe	**3**	**M2** for [2 ×] $\frac{16}{80}\times\frac{14}{79}$ oe or **M1** for $\frac{16}{80}$ or $\frac{16}{79}$ oe or $\frac{14}{80}$ oe or $\frac{14}{79}$ oe seen If 0 scored, **SC1** for answer $\frac{7}{100}$ oe
3(a)(i)	$\frac{AD}{46.1}$ = tan 64 oe or better	**M1**	
	94.51 to 94.52	**A1**	
3(a)(ii)	46[.0] or 45.96… nfww	**3**	**M2** for $56.5\times\frac{\sin 94}{78.4}$ oe or **M1** for $\frac{56.5}{\sin BAC}=\frac{78.4}{\sin 94}$ oe
3(a)(iii)	102.3 or 102.4 or 102.34 to 102.38	**4**	**M2** for [cos$C$ =] $\frac{38.6^2+78.4^2-94.5^2}{2\times 38.6\times 78.4}$ or **M1** for $94.5^2=38.6^2+78.4^2-2\times 38.6\times 78.4\times\cos C$ and **A1** for –0.214 or –0.2144 to –0.2137 If 0 scored, **SC2** for [CAD =] 23.5 or 23.51 to 23.52 or for [CDA =] 54.1 or 54.14…
3(b)	16.2 or 16.15…	**3**	**M2** for $\frac{1}{2}\times 21.5\times 27.6\sin 111=\frac{1}{2}\times 34.3\times d$ oe or **M1** for $\frac{1}{2}\times 21.5\times 27.6\sin 111$ seen or $\frac{1}{2}\times 34.3\times d$ oe soi
4(a)(i)	Image at (–5, 6) (–5, 8) (–6, 7)	**2**	**B1** for translation by $\begin{pmatrix}-4\\k\end{pmatrix}$ or $\begin{pmatrix}k\\5\end{pmatrix}$
4(a)(ii)	Image at (3, 1) (3, 3) (4, 2)	**2**	**B1** for reflection in $y = 1$ or $x = k$
4(a)(iii)	Image at (3, 4) (3, 8) (1, 6)	**2**	**B1** for enlargement, sf 2, in wrong position
4(b)	Rotation 90º [anticlockwise] oe (–3, 0)	**3**	**B1** for each